

Potassium Dichromate



What is it?

Chromium is a metal that can be used in steel alloys and can for corrosion resistance. It can also be used for green, yellow and red pigment, as well as for wood preservation and tanning leather. Chromium allergy is tested with the chemical Potassium Dichromate.

Chromium dermatitis is known for being thick and long standing. The rash may persist long after stopping contact with chromium.

POTASSIUM DICHROMATE can also be known as:

- Chrome and chromate
- Chromite and chromium
- Hydrated chromium sesquioxide
- Dichromium trioxide
- Related compounds: Ammonium dichromate, Chromated zinc chloride, Chromic acid, chromic nitrate, Chromium dioxide, Chromium hydroxide ultramarines, Chromium phosphate, oxide, or potassium sulfate, Potassium bichromate, Sodium dichromate, zinc chromate

Some Possible Uses of POTASSIUM

DICHROMATE / Where it might be found:

- Cosmetic pigments
- Leather, tanned (i.e. in gloves, shoes, garments, purse, belt, suede, etc)
- Wet cement, mortar, and brick sand
- Face scrub or mask
- Soap bars (e.g. Irish Spring)
- Shaving cream (can prevent canister rust)
- Matches
- Ceramic and artist pigments (green yellow)

- Tattoos (less common in modern tattoos)
- Textile dye fixing agent (e.g. military green)
- Roulette and pool table green felt
- Shoe polish, floor wax
- Chromic catgut and chromic silk suture
- Orthopedic prostheses
- Acupuncture needles
- Orthodontic wires, dental fillings, implants
- Chrome nickel dental crowns
- Industrial detergents, esp. phosphate
- Photo chemical, photo paper, blueprints
- Engraving, lithography, and offset
- Magnetic tape
- Epoxy filler and spackle
- Drywall, primer, paint
- Pressure treated wood
- Anti-fungal in grout, veterinary anti- fungal
- Gun and carburetor degreaser
- Etching agent, electro engraver of copper
- Metallurgy, electroplating, galvanizing, and welding
- Antioxidant for anti-freeze, radiators, and pipelines
- Anti-corrosive in dip, oils, fuels, drilling muds
- Explosives, fireworks, safety match heads
- Glues (i.e. animal hide glues)

Other products that might react like

POTASSIUM DICHROMATE: None

What should I do to avoid POTASSIUM DICHROMATE?

Avoid using any product that has chromate. It is important to check the ingredient list of each product you use. Some product ingredient lists

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are available on the internet at the manufacturer's or store's website.

Replace leather shoes/boots with vinyl (PVC), rubber, canvas or synthetic, or purchase custom-made shoes/boots. Always double check before purchase because not every shoe by a company may be chromium/leather-free.

- Boots: Wesco Boots (wescoboos.com); Multnomah Leather Shop (multnomahleather.com); Loveless Shoes (lovelessshoesandboots.com)
- Sandals: Crocs; Birkenstock; Juju Footwear; Jellies
- Casual: Beyond Skin (veganstyle.com.au); TOMS; SeaVees; Some Keds
- Gym: Allbirds; Some Saucony
- Work: Servus Injection Molded Footwear

Or, use aldehyde or vegetable tanned leather: mooshoes.com or zappos.com (choose vegan option).

Instead of leather gloves, wear heavy duty fabric, canvas, rubber, or vinyl. For dental crowns, use porcelain. For prostheses, consider titanium.

Problems may come from contact with chromium fumes. This may occur in processes like welding or electroplating. Try to use an exhaust hood, respirator, or other means of venting or shielding.

Jobs that may encounter chromium:

- Construction workers
- Foundry workers (added to sand for bricks)
- Makers of acetylene, aniline, artificial flowers, batteries, bleachers, blueprints, and candles

- Marble colorers, color makers, fabric dyers
- Frosters (e.g. glass and pottery)
- Furniture and glass polishers
- Ink makers
- Linoleum workers
- Offset printers
- Paper, pencil, and crayon makers
- Photography
- Rubber and tannery workers, upholsters
- Textile printers
- Vulcanizers, welders, and wood workers
- Cleaners of guns and carburetors
- Makers of fire extinguisher components
- Veterinary workers

To find a possible workplace exposure, check the Material Safety Data Sheet (MSDS). Check all solutions and products that you encounter.

OTHER:

Incorporation of iron sulfate into cement may help prevent a reaction.

Trace chromium may be present in cosmetics as a contaminate and not listed on the label. Your doctor may recommend a repeat open application test (ROAT) to determine if cosmetic products are safe.

Reference: *Dermatitis* 2019;30:87-105.

In addition, the ACDS Contact Allergen Management Program® (CAMP®) can help manage contact dermatitis. It helps to find products that are safe to use by creating a list of products that do not contain one's allergens. For more information about CAMP®, please ask your doctor or visit www.contactderm.org.

REVISION DATE: NOVEMBER 2025

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